

Tejas Kale

Senior Data Scientist

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I am a senior data scientist with 11 years of experience architecting and deploying scalable machine learning solutions to solve critical business problems for internal and external stakeholders. I have a proven ability to deliver significant business value (annual savings of €1m) by leading the full lifecycle of data science projects, from translating client objectives into technical solutions to final production deployment.

WORK EXPERIENCE

ALDI DX • Mülheim an der Ruhr, Germany 05/2024 - Present Senior Data Scientist

- Drove the end-to-end implementation of a machine learning solution to solve complex staffing challenges at ALDI Austria, developing a demand-driven labour forecasting model that directly informed staffing strategy, empowering store managers to enhance staffing efficiency, boost employee satisfaction, and deliver a predicted annual saving of 1m€.
- Developed a labour curve visualisation dashboard in Tableau, saving the organisation €500k in subscription fees for existing store management software.
- Improved logistics analytics by designing a labour budget algorithm, leveraging warehouse delivery times to achieve a ~25% reduction in inefficient labour allocation
- Collaborated with cross-functional client stakeholders to translate business objectives into ML solutions, effectively communicating complex findings to accelerate the project lifecycle and reduce deployment time by 3 months.
- Led hypothesis-driven analysis of store operations, for instance, identifying discrepancies in stocking practices – including warehouse delivery – and proposing iterative improvements to labour curve development especially for FMCG and special goods.

Relay GmbH • Berlin, Germany 02/2022 - 03/2024 Data Scientist

- Owned the end-to-end development of a predictive maintenance system, from data analysis and model training (Python, scikit-learn) to deploying and maintaining the prediction service in a cloud environment, resulting in a >90% reduction in critical equipment downtime.

SKILLS

ML & Data Science

- Azure
- Databricks
- Generative AI
- NLTK
- PySpark
- Python
- SQL
- Tableau

Non-Technical

- Agile
- NLP
- Orchestration
- Reproducible Research
- Stakeholder Management

- Developed, deployed, and maintained the analytics pipeline written in Python and Spark to monitor the performance of rotating equipment. Also implemented model improvements using Numpy, scikit-learn, Pandas, and MLflow.
- Developed and deployed a robust MLOps system based on Azure to monitor the performance of ML systems like detecting data drift through Log Analytics and triggering re-training based on events using Azure Functions.
- Developed and maintained Databricks dashboards for product management and executive leadership, identifying redundant cloud resources – resulting in annual cost savings of €12k through anomaly detection and root cause analysis.
- Improved the existing feature generation model (linear regression) for health state classification of rotating equipment used in industrial manufacturing which reduced false positive rate by 80%.
- Engineered a Spark-native analytics pipeline for monitoring rotating equipment, applying software engineering best practices to reduce execution time by 30% and cut developer debugging time by 90%.
- Developed and maintained Databricks dashboards using SQL and Unity Catalog (a relational database in Databricks) that reduced the asset health evaluation time required by Vibration Analysts by 60% and was awarded a Standing Ovation award by the team for the same.

Cumulus Systems Pvt. Ltd. • Pune, India **05/2014 - 12/2021**

Lead Architect

- Established the data science and analytics department in the organisation responsible for researching and implementing analytics solutions for business use-cases. Provided functional leadership for a small cross-functional team, mentoring and supporting other data scientists in their professional development.
- Developed anomaly detection models for flagging fraudulent credit card transactions using deep learning with TensorFlow (VAE) and boosted trees (LightGBM) which had a precision of 0.75 and recall of 0.9.
- Developed a NLP-based clustering and classification model for compression of system logs which reduced their storage requirement by 75%.
- Engineered an end-to-end computer vision (CV) pipeline using an InceptionV3-based architecture in TensorFlow to detect diabetic and hypertensive retinopathy, achieving a Quadratic Weighted Kappa score of >0.85 on a 2-class problem despite being restricted to a limited dataset of less than 10,000 lower-quality images.

Inter University Centre for Astronomy and Astrophysics • Pune, India

01/2012 - 03/2014

Statistical Programmer

- Wrote the statistical backend of VO-I's tool AstroStat in R.
- Conducted in-house and external workshops explaining the scope and usage of statistical analysis and VO-I tools in Astrophysics research.

EDUCATION

M.Sc. in Applied Statistics
University of Oxford • Oxford, UK • GPA: 3.0/4.0 10/2010 - 09/2011

B.Sc. in Statistics
University of Mumbai • Mumbai, IN • GPA: 3.4/4.0 07/2007 - 06/2010